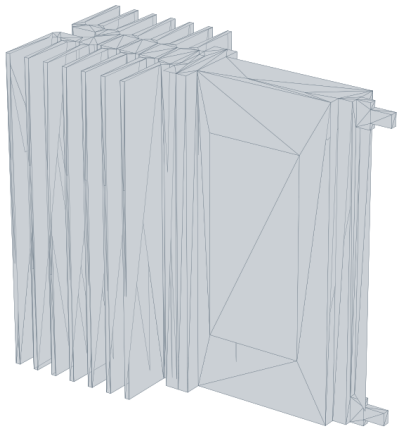
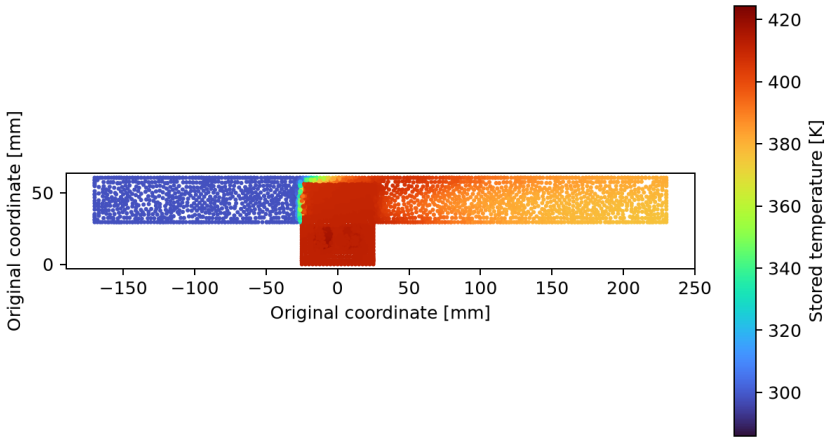
Radiator assembly: native CAD and saved FEM

13 October 2026. This extension adds two repositories with three original CAD geometries and seven saved analysis candidates: two Elmer cooling models and five wheel-rim variants. A previously collected test frame is identified as a duplicate; four unmatched strip restart outputs remain held. No new CAD or solver run.



Read-only original BREP view: four TO-220 transistor cases on an aluminium radiator. Native full-bridge_on_radiator.FCStd contains a valid 24-solid, 355-face final geometry: 23 physical solids plus an air domain. Display hides only the original air solid; no body was redesigned or reconstructed.



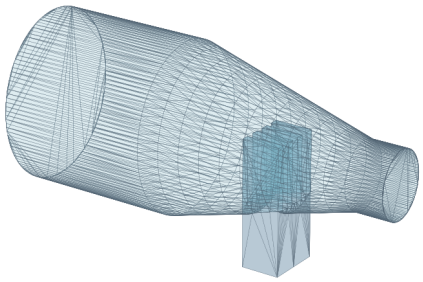
Read-only section of the existing saved Elmer field: 137,182 points / 130,305 cells. Original global temperature range is 285.954701 to 424.355500 K; pressure -0.01161098 to 0.05898355 Pa; derived speed maximum 0.18218990 m/s. Section displays original points within 1.5 mm of the median X plane; full arrays remain unchanged.

Native inlet/ambient temperature is 298.15 K and flow velocity 50 mm/s = 0.05 m/s. The global minimum is below ambient; finite arrays do not establish a physically validated solution. Keep the undershoot unmodified and hold scoring until mesh, stabilization, domain and convergence checks resolve it.

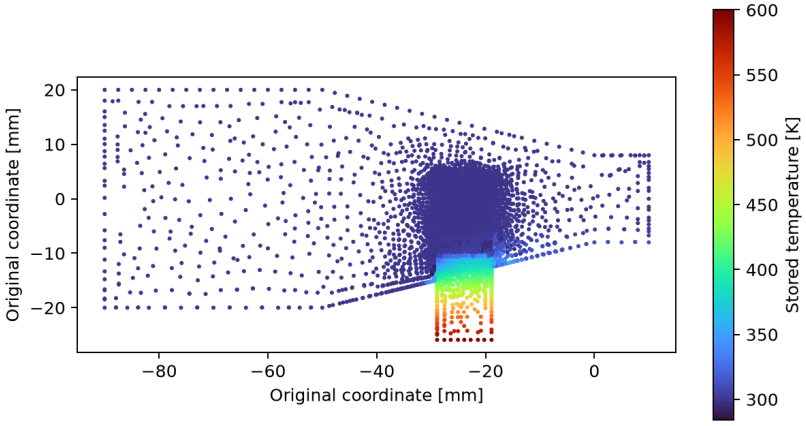
Publisher README heat powers 3.04 / 1.91 / 4 / 0 W total 8.95 W. Native DFlux values on three 9 mm2 faces imply 3.4 / 4.7 / 2.2 W total 10.3 W under Elmer SI flux units. This is a read-only derived load check; exact revision/scope disagreement is retained. Radiation is omitted in the source model.

Source: original publisher repository. Original bytes, source revision and file hashes retained.

Water cooling: coupled thermal-flow control



Read-only source CAD: native fem-elmer-tube-thermalflow-velocity-3d.FCStd stores two valid solids,29 faces and envelope 100 x 40 x 45.9 mm. Source steel block and water domain are shown, with the original fluid solid transparent. This is a simple coupled-physics control, separate from complex assembly counts.



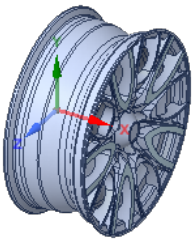
Original saved quantity	Native field / source setup
Mesh	41,565 points / 36,843 cells; full VTU and exact numerical copies
Temperature	284.267569 to 600 K; source 300 K inlet and 600 K imposed heated faces
Pressure	0 to 101,095.060167 Pa globally; zero solid-point values do not establish minimum fluid pressure
Velocity	Derived speed maximum 0.49931105 m/s; prescribed inlet 50 mm/s = 0.05 m/s
Material cards	Steel k 43 W/m/K; water k 0.591 W/m/K, rho 998 kg/m3, mu 0.001003 kg/m/s

The 600 K maximum is prescribed. The 284.2676 K minimum falls below the300 K thermal boundary values; do not clip it or treat it as a certified temperature target. Native model, mesh, full stored fields and finished solver log are included, but model convergence and domain-scoped scoring still need review.

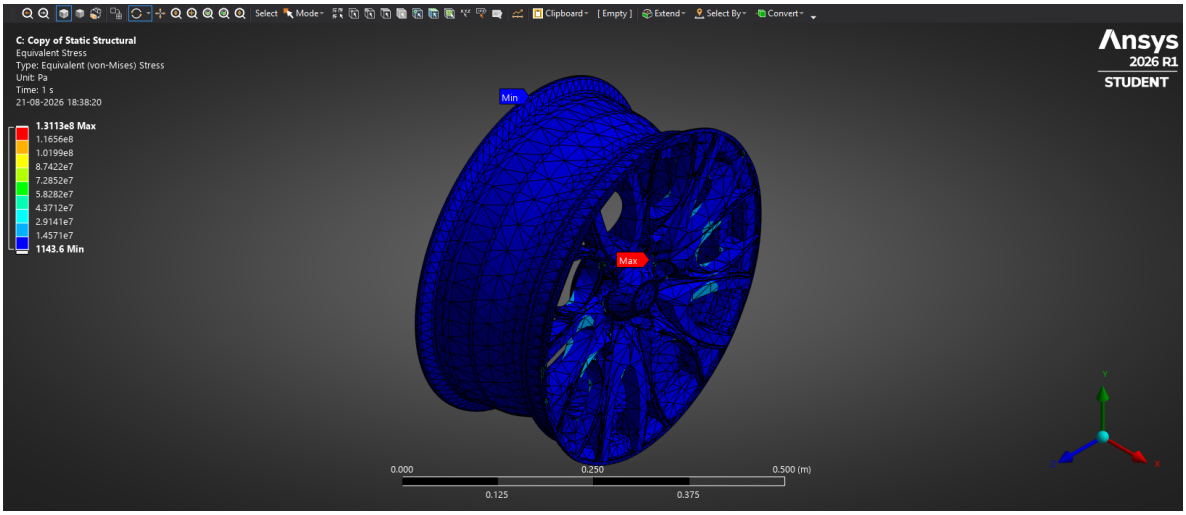
FreeCAD 0.21.1 Elmer writer uses SI scalar fields and scales mesh coordinates 0.001 during solving, reverting coordinates for VTU output. Thus stored coordinates correspond to original CAD millimeters, while temperature, pressure and velocity use K, Pa and m/s. Primary unit implementation. VTU lacks body/cell IDs; original UNV groups and native material references are retained for future mapping.

Source: original publisher repository. Original bytes, source revision and file hashes retained.

Wheel rim: native CAD and original static results



Unchanged original SpaceClaim thumbnail: native Rim.SLDPRT and Rim.scdocx included, with three duplicated native CAE imports in the project. Compatible independent solid import is pending. This detailed spoke/flange/hub geometry is one family, irrespective of five analysis variants.



Original bending FEM figure: displayed von Mises maximum 1.3113e8 Pa = 131.13 MPa, consistent with the report 131.1 MPa. Read-only averaged tensor recovery gives 131.132528 MPa. Original radial figure maximum 2.0905e7 Pa agrees with recovered 20.905141 MPa.

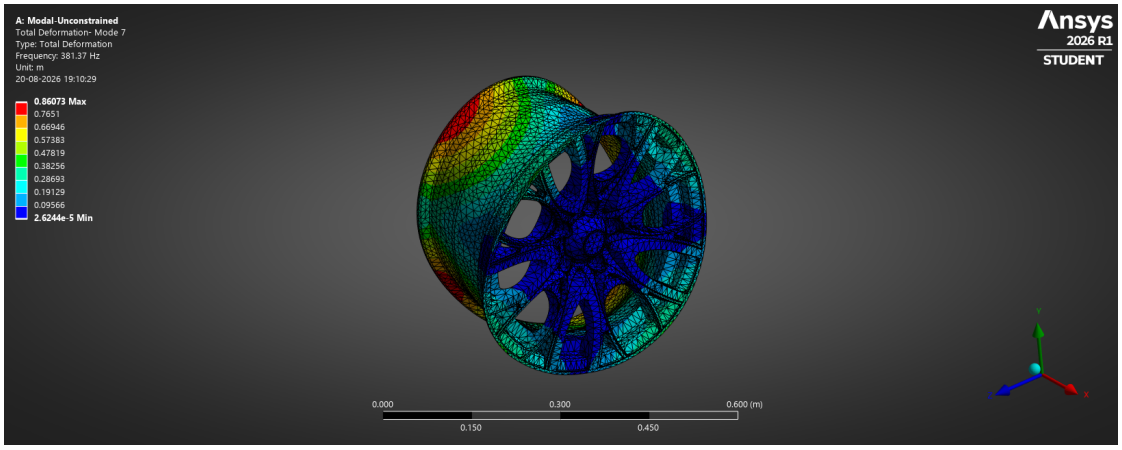
Source static case	Published U /stress	Saved references
Radial load	26.16 micrometers / 20.9 MPa	Source remote FY= 4000 N
Bending moment	244.3 micrometers / 131.1 MPa	Source MX= 1525, MY= 1296, MZ= 0 N m
Both static files	58,872 nodes / 32,734 elements each	9,378 finite stress nodes; 49,494 missing, masks retained

Source /units,MKS: E 71 GPa, rho 2770 kg/m3, nu 0.33; report yield 280 MPa. The report rounds a formula giving 1980 N m to 2000 N m; source component moment cards are preserved exactly. Static loading is a snapshot, not a measured fatigue life or reproduced qualification test.

Decoder limit: ansys-mapdl-reader 0.56.0 returns nonfinite translations and values near 1e308 for these 2026 R1 RSTs. Displacement arrays are excluded from NPZ. Native RSTs and publisher U plots remain unchanged; averaged stress arrays are exploratory pending compatible independent recovery.

Source: original publisher repository. Original bytes, source revision and file hashes retained.

Wheel modal results: three boundary variants



Original free-free Mode 7 figure: first flexible mode near 381.37 Hz. Six preceding rigid-body modes are retained from native headers, including numerical frequencies 0 to 0.00251 Hz. Mode deformation normalization is arbitrary; it is not an operating displacement.

Original saved model	Nodes /elements	Selected header frequencies [Hz]
Free-free, SYS	116,074 / 63,727	7-8: 381.371750 / 381.378935; 9-10: 771.759216 / 771.800400
Bolt-constrained, SYS-1	58,870 / 32,440	1-2: 390.017006 / 390.215152; 3-4: 433.718078 / 443.721897; 7-8: 780.500409 / 780.550708
Tire approximation, SYS-2	58,882 / 33,913	5-6: 20.843524 / 20.843526; 7-8: 283.770219 / 283.953610; 9-10: 613.176559 / 613.213384

All 12 frequencies per configuration are copied exactly into NPZ. Three original modal RSTs and two static RSTs have distinct SHA-256 values. The 36 modes count as three modal cases on one CAD family, not 36 new engineering drawings or geometries.

Constrained model fixes five cylindrical bolt-hole faces. Tire extension uses10 kg distributed mass and four 50,000 N/m radial springs; it is not a full 3D tire/contact model. Source decks, original geometry imports, material cards, plots and report are included.

Mesh discrepancy: report claims a converged 116,074-node mesh for all cases, but native constrained/tire/static headers contain about 58,870 nodes. Preserve each actual mesh. The published free-free convergence study does not certify all other saved cases. Report wording claims a shift below 0.2 Hz for 771.8 to 780.5 Hz, which does not match those numbers; use original frequencies.

Exact native geometry import, BC/recovery validation and mesh revision mapping remain pending. No verified complete dimensioned manufacturing drawing was located. No model trials or certified scoring tolerances.

Source: original publisher repository. Original bytes, source revision and file hashes retained.

Test frame duplicate and unmatched strip outputs

Original Test Frame.FCStd is byte-identical to the frame already collected in the earlier assembly batch. Its 2,356-node embedded result is not counted again. The newly inspected test_runs folder supplies four 316-node CalculiX 2.19 FRD displacement files and corresponding reaction totals.

Original stage	Applied UX /UZ [mm]	Reaction FX /FZ [N]
1	0.008723224 / 0.195083992	1532.754 / 9.899221
2	0.143245675 / 0.891315677	25170.38 / 45.19469
3	0.280735514 / 1.653076937	49329.41 / 83.81534
4	0.156580863 / 1.564182474	27513.38 / 79.34368

Original files: DAT totals are forces at the prescribed-displacement node set. The fixed-set totals oppose them to the stored rounding; XY/Z components and residuals are retained in the JSON audit. All 316-node displacement arrays are finite and copied with original node IDs and coordinates.

Geometry mismatch: each FRD has bounds 600 x 50 x 10 mm, while native frame bounds are2000 x 1500 x 100 mm. The provided CAD is not the matching strip geometry. No native strip CAD was located, and none was constructed locally. Paired new-family count 0.

The source decks use sequential *Static restart steps with steel E 210,000 MPa, nu 0.3. Labels1..4 are load stages, not a dynamic digital-twin trajectory. Only DISP is requested; these files do not contain a stress field. Maximum edge displacement can be prescribed input, so reaction totals are the independent response candidates after geometry matching.

Four stages remain one unmatched strip family/control lead, not four complex CAD/FEM tasks. Original restart and solver-output files are retained; vendored executables, manuals, FMUs and unrelated co-simulation runs were excluded. No source program or solver was executed.

Source: original publisher repository. Original bytes, source revision and file hashes retained.

Inventory, provenance and benchmark readiness

New original paired evidence	Count / qualification
New repositories /CAD geometries	2 / 3: radiator assembly, water-cooling control, wheel rim
Complex geometries /simple controls	2 / 1
Saved analysis candidates	7: 2 Elmer coupled fields +5 wheel static/modal files
Held/duplicate testbench	Duplicate frame adds 0; 4 unmatched strip restart stages
New dimensioned drawing targets	0 verified
Certified admissions /frontier scores	0 /none

Verification: 178 original downloaded files pass pinned Git blob SHA-1, byte size and SHA-256. Three FCStd archives pass CRC and byte comparison across 701,63 and 84 members. Two full Elmer VTU copies match all numerical arrays and cell connectivity exactly. Static wheel stress copies match repeat reads, including NaN masks; five RST hashes are distinct.

Supplied CAD to FEM: original Elmer cached BREPs independently import as valid solids. Native wheel CAD needs compatible import. Geometry/material/body assignments, dimensional scaling, heat-flux scope, fixed/load faces and solver/recovery conventions must be frozen before numerical grading. Finite raw arrays and finished logs alone do not establish physics accuracy.

Drawing to CAD to FEM: keep a separate track. CAD thumbnails, cached BREP views and result figures are included, but no complete matching dimensioned drawing was verified. No source drawing or missing CAD was synthesized. Nodes, time labels, mode numbers and duplicated project imports do not inflate item counts.

Target limits: retain Elmer temperature undershoots and the radiator README/native heat-load disagreement. Do not score prescribed 600 K as a prediction, interpret solid-node zeros as fluid pressure minima, use normalized modal shapes as static deformation, or accept invalid legacy displacement decoding. Wheel convergence claims require exact case/mesh mapping.

Other crawled leads: heated Inconel duct repository has reports, CSVs and decks but explicitly omits solver binaries/nativeCAD; CubeSat source has tensile experiments without CAD; robot-dog source has complex CAD without FEM outputs; sector gate has reports/video without native CAD; NIC shroud has analytical tables and generation/solver scripts without uploaded native CAD/full fields. None adds a paired item.

Package policy: original FCStd, native wheel CAD, five RSTs, solver decks/logs, plots/reports, extracted VTUs, exact numerical copies, catalogs and master snapshot included. Original FCStd files contain every cached geometry/mesh member; duplicate extracted history members are omitted. Selected final BREP, Document.xml and native solver text accompany extracted fields.

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